

# A Risk Analysis Framework for Cyber Security and Critical Infrastructure Protection of the U.S. Electric Power Grid

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**ABSTRACT:** The purpose of this article is to introduce a risk analysis framework to enhance the cyber security of and to protect the critical infrastructure of the electric power grid of the United States. Building on the fundamental questions of risk assessment and management, this framework aims to advance the current risk analysis discussions pertaining to the electric power grid. Most of the previous risk-related studies on the electric power grid focus mainly on the recovery of the network from hurricanes and other natural disasters. In contrast, a disproportionately small number of studies explicitly investigate the vulnerability of the electric power grid to cyber-attack scenarios, and how they could be prevented or mitigated. Such a limited approach leaves the United States vulnerable to foreign and domestic threats (both state-sponsored and “lone wolf”) to infiltrate a network that lacks a comprehensive security environment or coordinated government response. By conducting a review of the literature and presenting a risk-based framework, this article underscores the need for a coordinated U.S. cyber security effort toward formulating strategies and responses conducive to protecting the nation against attacks on the electric power grid.

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**KEY WORDS:** Electric grid; failure mode effect analysis; hierarchical holographic modeling; risk analysis; risk filtering

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## 1. INTRODUCTION

The electric power grid is a complex interconnected system. The electric power grid also is an element of the “critical infrastructure and key resources” (CIKR) according to the National Infrastructure Protection Plan (NIPP, 2017). Electricity is a requirement for daily life in today’s world. Modern society could not function without a reliable source of electricity. Without electric power, computer systems, banking systems, and fuel systems shut down, leaving most of the population with no means to power their household, most businesses unable to

run, and the government unable to provide services. Berst (2018) noted, that in the United States alone, the change from an agrarian society to a modern industrialized nation was due in large part to electricity, as evidenced by the global electricity demand of 2.06 terawatts (TW) in 1950 versus 3.8 TW in 2000, and a predicted demand of 6.99 TW in 2050. Today, we depend on electricity for basic needs, such as food, water, shelter, communication, employment, and healthcare.

However, the subject remains largely underdeveloped and unexplored. Many of the current studies recognize the need for a quick recovery of the electric grid but treat it as a regional problem that can easily be tackled using resources from neighboring states to recover operations efficiently (Stockton, 2016). For example, Hurricane Matthew in 2007 demonstrated that natural disasters can debilitate the network and impair recovery efforts—it left

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approximately 1.4 million people without power (Chediak & Stillman, 2016). Yet, soon after the storm, efforts were underway to restore the grid in the affected areas using assistance from neighboring states and other resources. More recently, Hurricane Maria and its impact on the island of Puerto Rico highlight the slow recovery of the power grid on the island and the inefficient response to this event (Vick, 2017). The U.S. economy is inherently dependent on electric power for all of the important aspects of life. The Northeast Blackout of 2003 revealed that an outage can have far-reaching effects on other critical infrastructure systems (Anderson, Santos, & Haimes, 2007).

A recent Federal Emergency Management Agency (FEMA) Private Sector Call Summary (NBOEC, 2016) estimated that 90% of utilities across the impacted states of Florida, Georgia, South Carolina, North Carolina, and Virginia (Hurricane Matthew) would be restored after only seven days. In contrast, a cyber-attack on six power substations in Ukraine in December 2015 thrust 235,000 people into the cold and dark (Lee, Asante, & Conway, 2016). This event only ceased when coordinated efforts were implemented to mitigate the attack.

Kaplan and Garrick (1981) offered a fundamental definition of risk that can be applied to a risk analysis of both natural disasters and a cyber-attack on the electric grid. In addition, Haimes (1991) offered a structure to compare natural disaster events and cyber-attacks against the electric grid within a risk management framework.

Kaplan and Garrick (1981) posited a structured risk analysis using the following triplet of risk assessment questions:

- (1) What can go wrong?
- (2) What is the likelihood?
- (3) What are the consequences?

Haimes (1991) supplemented the above questions with a subsequent triplet of risk management questions:

- (1) What can be done and what options are available?
- (2) What are the trade-offs among costs, benefits, and risks?
- (3) What are the impacts of current decisions on future options?

This article builds on the above risk assessment and management questions, developing a discussion

given the current literature on the subject, and compares and contrasts the similarities and differences of a natural disaster grid outage versus a cyber-attack. The subsequent discussions describe various risk scenarios (e.g., natural disaster or cyber-attack scenarios), the likelihoods of occurrence, and their consequences. Subsequently, the authors examine the risk management process using the above triplet of questions by Haimes (1991) to determine the response required and if it differs between the two events and how current decisions are impacting future options.

Key research questions that guide the selection of articles for the literature review are as follows:

- How would the response to a cyber-attack be different from a natural disaster?
- What lessons can be learned from previous cyber-attacks on other sectors and regions?
- How plausible is a cyber-attack to an electric power grid?
- What are the direct and indirect consequences of an attack to the electric power grid?

The organization of this article is as follows. Section 2 presents a comparative risk analysis of the electric power grid for natural disasters and cyber-attack scenarios. Section 3 performs a review of methods relevant to the study, notably the Hierarchical Holographic Modeling (HHM), the risk filtering and risk management (RFRM), and the failure mode, effects, and criticality analysis (FMECA). In Section 4, we begin the application framework to prioritizing risk scenarios pertaining to the electric power grid. In Section 5, RFRM allow us to qualitatively and quantitatively examine the system using the eight phases of the process. Finally, Section 6 will offer conclusions and opportunities for future use of this work.

## 2. RISK ANALYSIS OF ELECTRIC POWER DISRUPTIONS

This section presents a comparative risk analysis of the electric power grid for natural disasters and cyber-attack scenarios, which will serve as the overarching motivation for the development of the proposed risk analysis framework.

### 2.1. Risk Assessment of Natural Disaster Disruptions

Natural disasters have increasingly become part of yearly weather cycles, which for many areas cause

major disruptions to citizens, businesses, and governments, due to electric power outages. Many regions of the United States are vulnerable every year to extreme weather conditions, such as flooding along the Mississippi River, tornadoes in Oklahoma and Kansas, flooding, mudslides, earthquakes, and forest fires in California, and hurricanes in the Gulf of Mexico and the Central Atlantic region of the United States, among others. In each of these situations, many areas lose critical services due to catastrophic power failures.

### *2.1.1. What Can Go Wrong?*

The economic impact of a failure of the electric grid on a massive scale cannot be overstated. Before exploring the cyber-security risks of the electric power grid, it is useful to first delve into the analysis of grid vulnerability to natural disasters, like hurricanes. In the case of Hurricane Katrina, the United States lost more than \$200 billion; furthermore, the recovery time increased significantly because of the lack of electricity (Bureau of Labor Statistics, 2017). Grid outages have a profound and lasting effect on the people and economy of the affected area because of the lack of power. Several disaster risk management and economic models delineate the direct and indirect effects of grid outages due to natural disasters. For example, historical occurrences of widespread network outages occurred in New York City in the 1960s and 1970s, affecting over 34 million people (McFadden, 2013). Anderson and Santos detailed another relevant scenario in the Northeast blackout that occurred in 2003. The paper details the regional and far-reaching effect of a natural disaster and the potential consequences and duration of such events. The event affected New York, Pennsylvania, Ohio, Michigan, and the province of Ontario, which was triggered by a software bug. In addition, Eusgled and Kruger detail the interdependent nature of critical infrastructures through modeling and simulation techniques. Finally, there has been significant work done by Santos and Herrera et al. on the workforce criticality influencing disaster preparedness. Disaster models have shown the crippling effects of the lack of power and subsequent loss of potable water and proper refrigeration, and the negative impact on the Gross Domestic Product (Department of Energy, 2015). Modern society depends on energy services. Without electricity, homes and businesses are left in the dark, which leads to costly business interruption losses (Sandalow, 2012).

### *2.1.2. What is the Likelihood?*

Hurricane season in the United States lasts from June 1 to November 30. Approximately 12 tropical storms, six of which become hurricanes, form over the Atlantic Ocean, Caribbean Sea, or the Gulf of Mexico each year (National Weather Service, 2016). Storms such as Katrina and Rita leave many millions without electricity, and the recovery often lasts for weeks (Farber-Deana, Décor-Souza, & Laramie, 2009). International storms like Typhoon Haiyan (which devastated the Philippines and is one of the strongest tropical cyclones on record) can bring “massive destruction with thousands in fatalities destroying the electric power networks and making the rescue and recovery operations extremely challenging” (Santos, Herrera, Yu, Pagsuyoin, & Tan, 2014). The consequences of hurricanes that make landfall in the continental United States are relatively minimized and brief. With the exception of a Hurricane Katrina scenario, damage is limited to a small region with power outages numbering in the days. The Hurricane Matthew storm repeated this model as the majority of the power was restored within a week.

### *2.1.3. What are the Consequences?*

The implications of power outages due to hurricanes are relatively small. It is a well-practiced scenario with all but the most intense storms allowing recovery within a week. The practiced recovery model is built on the case of an isolated sector, such as the Gulf Coast, being reinforced by forces from surrounding states assembling and aiding in the recovery. This model allows the forces necessary to focus on the target, because of previously developed memoranda of agreement (MOA) that call for reciprocity among the affected states. Typically, the affected State’s National Guard is activated, and the FEMA brings federal government assistance to bear. This scenario is plausible because of the relatively small area to control. In addition, much has been accomplished with the advent of the Internet. Local electric concerns, like the Potomac Electric Power Company (PEPCO), have applications for smartphones that detail the time of the outage, population affected, and an estimated time for the return of service (PEPCO, 2016). Models of multiple paths for the hurricane typically quickly converge (NOAA, 2009), allowing forecasters to warn individuals likely in the storm’s path and for the government to prepare the population for possible evacuation.

Superstorm Sandy proved an appropriate case study of the current recovery model. The storm, while devastating, allowed for a quick recovery that is now considered a notable source of information for management of future events (Heslin, 2014). The question then becomes whether or not policymakers can manage cyber-attack scenarios by applying similar models used in the case of hurricanes.

## 2.2. Risk Assessment of Cyber Security Attacks

### 2.2.1. *What Can Go Wrong?*

The eventual impact of a cyber-attack scenario on the electric grid is similar to a natural disaster; namely, the outage of electric-power-dependent systems. Although natural disasters like hurricanes often directly damage the physical components of the electric power grid, cyber-attack scenarios could also significantly degrade the grid's capability to provide reliable electric power to households and businesses. Therein lie the similarity and the point of departure for the discussion of a cyber-attack. The example of the six substations in Ukraine going offline on December 23, 2015, in the dead of winter, is a prime example of a coordinated effort to simultaneously knock out power to a particular area. This incident has been attributed to a group affiliated with the Russian government (Siboni & Magen, 2016). It affected 235,000 people, who were left without electricity in December, when the average temperature throughout the month is just 28 °F (Fox-Brewster, 2016). A cyber-security attack could potentially pose as a more severe problem. Hackers could knock out key elements of the grid infrastructure. In addition, the hackers could anticipate the government's response and take out key infrastructure, like transformers, that can take many months to replace (Scribd, 2012). The motivation of the hackers is the most important element. Is it state-sponsored hacking, for a political end, or is it the act of holding the grid hostage for money? This latter example is similar to recent situations with hospitals (Krebs, 2016) where data were held hostage until a ransom was paid. This example involved a Kentucky hospital where hackers encrypted files through malware and held the files until a ransom was paid.

### 2.2.2. *What is the Likelihood?*

A typical hack remains on a network for an average of 205 days (Economist, 2015). The three

U.S. electric grids have discovered breaches within their networks (Gorman, 2009). Once resident on a system, these hacks can be triggered at any time and as often as desired. A cyber-attack could be triggered for ransom (Winton, 2016) similar to the recent malware attacks against hospitals, or the attack could be triggered by a larger, political situation. In a recent Harvard Business Review article, Madnick (2017) posited that there are the three conditions necessary for it to happen: opportunity, capability, and motivation. However, much like the 2007 denial of service attack in Estonia or the electric grid attack in Ukraine, these events are only the initial salvos in a war for which we must not fail to prepare. The United States, like many other developed countries, relies heavily on technology to fuel its economy and service its population, and technology needs electricity. Beginning with the National Research Council's *Terrorism and the Electric Power Delivery System* report in 2012 there has been a recognition of the need to restore power faster and make critical services less vulnerable. The President's National Infrastructure Advisory Council recommended a systems approach to plan and respond to these types of attacks. Finally, the DHS (2018) report detailed the resilience framework necessary to recover from such an attack. Recent work that has been accomplished in this area include the National Renewable Energy Laboratory's (NREL) research into energy security and resilience (e.g., potential service disruption consequences, researching scenarios caused by man-made or natural disruptions, and working toward the overall goal of more resilient energy systems). In addition, the Department of Energy has taken steps, through the Office of Cybersecurity, Energy Security, and Emergency Response (CESER), to strengthen preparedness, coordinate incident response, and achieve more resilient energy delivery systems. Finally, as reported in the Electric Grid Cybersecurity report by the Congressional Research Service (CRS, ), mandatory standards for the electric power grid may be an issue for Congress to consider.

### 2.2.3. *What are the Consequences?*

The effects of a cyber-attack on the electric grid are significantly different than a natural disaster event. The consequences could be more dire. Cyberattacks could affect the morale of affected populations and could cause a degradation of trust to the service providers (Farahani, 2016). While natural disasters are incapable of choosing a specific target (i.e.,

critical location or node of the network), a cyber-attacker can target a specific node, time, and method of attack, aided by information on the design of the grid, which are readily available in open source (nonclassified) material on the Internet (Cherdantseva, Bumap, & Blyth, 2016). Recently, Hurricane Matthew left an estimated 1.4 million people without electricity and killed over 1,000 people (Delva, 2016). While Hurricane Matthew was a devastating storm, the consequences could have been more dire with a slight change in the timing and trajectory of its landfall.

### 2.3. Risk Management of Cyber Security Attacks

Increased cyber exposure must be mitigated. With the advent of increased technology on the electric grid, there is greater vulnerability to cyber-attack. We have established the importance of the U.S. electric grid to the economy, and how vulnerable businesses and populations are with disruptions to the availability of electricity. Even more important is the risk management factors for such an event. Hence, it is of paramount importance to enhance the resiliency of the electric power grid against disruptive natural or manmade events, knowing that it is arguably one of the most critical lifeline infrastructure systems that directly supports the delivery of essential services and the realization of a myriad of economic activities (CRS, 2015). In an effort to address the question “What can be done?” the Office of Electricity Delivery and Energy Reliability (OE) delineates three major elements as part of a comprehensive approach to cyber-security. They include developing cyber-security standards and promoting public-private partnerships, while advancing risk management strategies (CRS, 2015). Such elements of risk assessment and management are considered to be important building blocks in the development of a comprehensive framework to cyber-security across the government.

Risk mitigation and risk management strategies need to be developed through a series of planning exercises at all levels of government, especially in the relatively nascent area of a response to a cyber-attack. Additional mechanical backups to computerized systems should augment systems that are at this point semiautonomous and represent a single point of failure. A key concept to consider is whether or not economic incentives are adequate (Dynes, Goetz, & Freeman, 2016). Backup systems, though costly, must be mandated to mitigate the effects of

such an attack. Financial incentives from the government could also entice companies to bring their organizations up to a common standard. Prioritization of recovery should be established, as well as prioritization of limited resources available during such an event.

Currently, a multitude of agencies, both state and federal, are active in this space. The typical response to a grid failure is *ad hoc*, with a hodgepodge of available agencies and resources assembling to address the problem. Though repairs are the same, the response to an attack must be different. Little has been done to codify the lessons learned or to develop a template for a response to a cyber event. Recent policy statements cite the need for a coordinated effort, but there remains no “lead agency” to respond to these incidents. There has been an increasing recognition of the importance of the public-private partnerships necessary to protect this fledgling area of cybersecurity in the context of electric power grids.

I, a unique capability rests with the National Guard units of each state (Department of Defense [DOD], 2017). The Guard has been trained to respond to cyber-attacks within their state and act cooperatively with other states. Each state has a Joint Operations Center (JOC) that acts as a centralized command post. Training has been standardized across the states and civilian skills (Guard members’ skills from their traditional day jobs) have been brought to bear against attacks. The federal government has developed cyber teams (from the aforementioned Guard units) to ensure that the respective state or regional electric concerns meet a basic standard and conducts penetration testing as a part of a series of exercises to determine the “weakest links” in the chain. This model is similar to the current thinking of U.S. Cyber Command, which has fielded 133 teams to protect and defend the Department of Defense networks.

### 3. REVIEW OF SUPPORTING MODELS

Most of the previous risk-related studies on the electric power grid focus mainly on the recovery of the network from hurricanes or other natural disasters. In contrast, a disproportionately small number of studies explicitly investigate the vulnerability of the electric power grid to cyber-attack scenarios and how they could be prevented or mitigated. This is evidenced by a simple search of relevant electronic databases, which reveal approximately a 4:1 ratio for articles about natural disasters and the electric



**Fig 1.** Framework for linking HHM, RFRM, and FMECA.

grid versus cyber-security and the electric grid. This compares with a sixfold increase in the number of cyber-attacks cited in a (DOE, 2016) report by the Idaho National Laboratory. By conducting a review of the literature and presenting a risk-based framework, this article underscores the need for a coordinated U.S. cyber-security effort, formulating strategies and responses conducive to protecting the nation against attacks on the electric power grid. This section presents a discussion of relevant material that forms the basis of the methodology used in this article. In a complex environment, such as quantifying risks to the electric grid from natural disasters and cyber-security threats, it is necessary to accommodate multiple stakeholders by developing many models of the system. In this article, we examine the electric grid system by first developing a HHM to understand the system under review. The icons in Fig. 1 show the progression of first identifying the risk scenarios in a system through the HHM. Subsequently, the risk scenarios identified via the HHM are qualitatively prioritized via the risk filtering, ranking, and management (RFRM). Finally, the prioritized risk scenarios are quantified using the FMECA to address the question of risk management.

The authors have considered a hybrid and mode-based approach as the paper logical line flows from the initial review of the problem set with hierarchy holographic modeling (HHM) to address the complexity and then to shape and scope the problem through the use of the RFRM and finally to “quantify the risk” by applying the FMECA. The scalability of this model depends on the overall scope of the problem that it is applied against.

### 3.1. Hierarchical Holographic Model

Haimes (1981) introduced the HHM to underscore the multiple, hierarchical, and overlapping coordination of issues and entities present in any sys-

tem. In contrast to most planar models, HHM aims to capture the multidimensional features of technological problems. It aims to identify risk scenarios by charting the critical issues in a complex system or organization. To construct an HHM, comprehensive documentation of the system under study is needed. Information-gathering techniques include interviews, surveys, independent audits and research, among others. The modeler needs to judge the importance of the identified issues, objectives, and visions of the system, and how they are interrelated. HHM provides a hierarchical structure of a system in different perspectives. It provides insights on the elements, objectives, and constraints that make up a system, which ultimately aid in the identification of present and potential risks. HHM is effective in decomposing, clarifying, and prioritizing sources of risks not only in the context of the system, but also to a vast number of economic and societal issues. The efficacy of HHM is more pronounced for systems that are large scale, complex and have interdependent subsystems, such as the electric power grid.

### 3.2. Risk Filtering, Ranking and Management

RFRM (Haimes, Kaplan, & Lambert, 2002) is the next step in our progression of managing a large set of risk scenarios for a system. It aims to filter the scenarios through the knowledge of the domain expert. This initial qualitative filtering and ranking is based on the likelihood and consequences of each risk. A quantitative ranking is then developed through the use of a matrix to define the scenarios with the greatest severity and likelihood of occurrence, considering the resiliency and robustness of the system. Next, a cost-benefit analysis is developed, and scenarios previously filtered out are reexamined. Each of the eight phases of the system is developed in further detail in subsequent sections of this article.

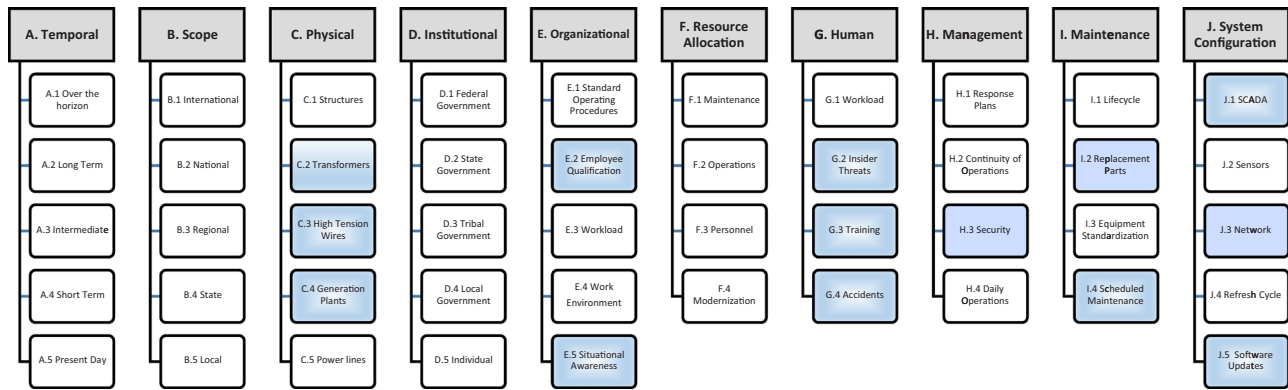


Fig 2. Hierarchical Holographic Modeling framework for electric grid.

### 3.3. Failure Mode, Effects, and Criticality Analysis

Failure mode and effects analysis (FMEA) and its extension, FMECA are powerful and effective analytical tools, which are widely used in engineering projects to examine possible failure modes and to eliminate potential failure during designs. In particular, it provides design engineers with practical quantitative or qualitative measures necessary to guide the implementation of corrective actions, by focusing on the main failure modes and their impact on products. Widely adopted by reliability practitioners, it has become the standard in various industries in Japan and America and in European manufacturing companies (Bouti & Kadi, 2012).

The risk priority number (RPN) is used to conduct the risk assessment. The three factors are all scored from 1 (best) to 5 (worst) on the basis of degree. RPN is the product of occurrence, detection, and severity, which is expressed as  $RPN = \text{Severity (SEV)} \times \text{Occurrence (OCC)} \times \text{Detection (DET)}$ .<sup>(46)</sup> (In various uses of FMECA, a scale of 1–10 is used, but it is arbitrary.) This simple, yet powerful formula allows us to rank the possible threats to the system and to develop a cost–benefit analysis and mitigation courses of action for specific scenarios.

## 4. APPLICATION OF FRAMEWORK TO PRIORITIZING RISK SCENARIOS

Managers and key stakeholders of the electric grid in the United States face a complex plethora of competing needs and requirements. The model described in Fig. 2 is the basis for our description of the problem. From this point forward, the integration of methodology will be demonstrated in a case study.

Each step of the RFRM has been validated in consultation with subject matter experts (SMEs).

### 4.1. Interviews

Subjects of these interviews were discovered through a purposive sampling. The SMEs represented senior officials with 10+ years of experience in this area. Their respective backgrounds included: government officials (to include a former undersecretary of homeland security), academia (represented by members of the Department of Defense and their associated military colleges), and members of key agencies in government to include both public and private entities involved in this area of study. The identities of the individuals interviewed are not revealed in this work to ensure nonattribution and the subsequent freedom of candor from the individuals interviewed.

### 4.2. Surveys

The purpose of the survey was to expand the initial group of experts to include another group in this study. This group numbered approximately 60 and represent individuals currently working in this field of study. The web-based survey was administered to these 60 individuals as a group separate from a greater population group that represented a cross-section of the population of the United States. This group numbered approximately 161 for a total of 221 respondents. A field testing was accomplished on the survey to see if any modifications were necessary for clarity and validity. No modifications were necessary. In addition, the commercial vendor of the

survey provided vetting of the questions before the survey was made public.

The response rate for the survey was approximately 85% for the expanded “expert” group. The cross-section group exceeded expectations as it was designed for 100 respondents, but 161 people completed the survey. The survey expanded from the initial group of experts in order to find the opinions of a population that will feel the effects of policy changes and fathom the awareness, education level, and expectations for a cyberattack on the electric power grid.

We approach this problem using the HHM model for events involving cyber-attacks given the input from both the structured interviews and survey responses (see Fig. 2). The framework for identifying sources of risk associated with the electric grid is defined by headtopics. Headtopics are chosen as an illustrative sample set of all possible “as planned” scenarios. These include temporal, scope, and institutional among others. These three previously listed headtopics are critical to any examination of the problem as they can be woven in and among the other headtopics and subcategories many times compounding the problem and complicating the results.

The 10 headtopics are defined below:

#### A. Temporal

The time horizon associated with a proposed scenario is critical to a project’s success. Time is key in the development of a cyber-attack as well as the response to a cyber-attack. The response to a cyber-attack must be in minutes vice the current number of days it takes to marshal the necessary equipment and manpower.

#### B. Scope

The scope is the defined areas or regions affected by the scenario. Many cyber-attack scenarios include a “cascading” effect as problems become more complex due to effects of one system being impinged or failing exerting a similar effect or connected systems.

#### C. Physical

The physical headtopic represents the important structures and specific unique equipment in the electric grid (e.g., generation, transmission, and distribution networks). Each system is arrayed around the fundamental building blocks of a process. These

physical structures are important in the everyday operation of the network and are obvious targets for attack.

#### D. Institutional

The key stakeholders and levels of ruling bodies associated with the grid come with competing interests as well as the ability to affect change and improvements. Oversight by various levels of governing bodies adds to the current challenge for successful operation of the electric power grid.

#### E. Organizational

The organizational problem area defines all incidents that are directly associated with the overall structure of the organization. This is probably the broadest and most encompassing area, as subtopics can range anywhere from the organizational boundaries such as vertical, horizontal, external, and geographic (Haimes et al., 2002), to the overall culture or work environment of the electric power grid.

#### F. Resource Allocation

Resource allocation involves the interplay of competing elements for limited resources. The problem area of resource allocation applies to all situations where there is a shortage of desired supplies or resources needed to accomplish a myriad of functions and processes encompassed by the electric power grid. This problem area is also defined by the potential side effects caused by eliminating resources.

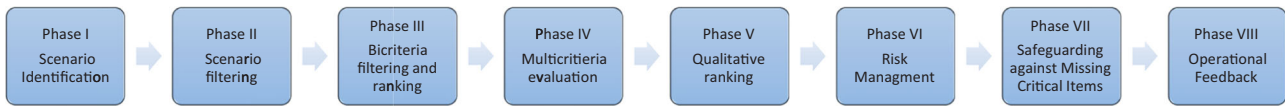
#### G. Human

The human element of a system cannot be underestimated or easily understood. It is both a strength and an important weakness of the electric power grid system. A human error is an individual action that significantly affects the success of a project. Human errors are generally classified as deviations from standard operating procedures (errors of commission) and skipping/overlooking necessary tasks (errors of omission). Insider human threats, such as those carried out by disgruntled employees fall within the human head topic.

#### H. Management

Management involves correctly and efficiently performing the functions and processes of an electric power grid. Project management establishes the budget and schedule for a project and oversees all development, reporting, training, and transition





**Fig 3.** Phases of RFRM.

**Table I.** List of 14 Scenarios to be Evaluated in Phase III

| Headtopics              | Subtopic ID | Subtopic Name         |
|-------------------------|-------------|-----------------------|
| C. Physical             | C.2         | Transformers          |
|                         | C.3         | High-Tension Wires    |
|                         | C.4         | Generations Plants    |
| E. Human                | E.2         | Employee Training     |
|                         | E.5         | Situational Awareness |
| G. Hardware             | G.2         | Insider Threat        |
|                         | G.3         | Training              |
|                         | G.4         | Accident              |
| H. Management           | H.3         | Security              |
| I. Maintenance          | I.2         | Replacement Parts     |
|                         | I.4         | Scheduled Maintenance |
| J. System Configuration | J.1         | SCADA                 |
|                         | J.3         | Network               |
|                         | J.5         | Software updates      |

procedures implemented throughout. Another responsibility embedded in project management is establishing well-defined work assignments and reporting hierarchies. It must also establish a co-operative environment between and among the organizational players and stakeholders.

#### I. Maintenance

Required maintenance, equipment lifecycle, and the replacement of parts are important elements of a maintenance program. Also, the necessary inventory is crucial in repairing systems whose components are tailored to the electric power grid system and may require inordinate amounts of lead-time in order to develop and fabricate.

#### J. System Configuration

The efficiency and security of the electric power grid element's configuration is vital to the overall success of the system's performance. This area encompasses pertinent functions and teams associated with the engineering and operation of the electric power grid to ensure that the key components are integrated precisely and operating at their optimal configuration. Oversights in this area could include failure to install software updates, misconfiguring the SCADA Supervisory Control and Data Acquisition-

devices, as well as failing to optimize sensors to alert for any possible intrusion or disruption to the system.

## 5. QUANTIFYING AND RANKING RISKS

The RFRM Framework allows us to qualitatively and quantitatively examine the system using the eight phases of the process (Fig. 3). The RFRM recognizes the complexity of the prioritization of the risk scenarios according to their criticality. Hence, it encompasses a multistage filtering process that involves qualitative and quantitative ranking. For cases where it is impractical to apply quantitative risk analysis to multiple sources of risk, qualitative risk analysis may be used to initiate the identification of critical risk scenarios, which would eventually serve as the basis for formulating and implementing risk management policies to mitigate the risks (Lamm & Haimes, 2002).

The eight phases of RFRM (see Fig. 3) represent a holistic approach for risk analysis, rather than a mechanical or sequential process. Here, ranking and filtering are viewed as precursors, rather than as substitutes to considering the risk associated with a given system. The aim of the risk filtering and ranking process is to provide priorities in the scenario analysis to explore more urgent sources of risk scenarios first (Haimes et al., 2002).

### 5.1. Phase I: Scenario Identification

This phase is accomplished through the construction of an HHM. This is a diagram that categorizes multiple perspectives of a system and encapsulates the gestalt of the sources of risk. An HHM describes an array of "as-planned" scenarios (see Fig. 2) which are represented by headtopics (i.e., perspectives) and are further decomposed to subtopics. HHM makes it easy to analyze system-specific risk scenarios (i.e., departures from "as-planned" scenarios).

### 5.2. Phase II: Scenario Filtering

This phase entails a process in which the risk scenarios initially identified are narrowed down into the

| Likelihood \ Impact | Unlikely | Seldom   | Occasional | Likely   | Frequent |
|---------------------|----------|----------|------------|----------|----------|
| Catastrophic        |          |          |            | C.2      |          |
| Critical            |          | I.2      |            | J.1; H.3 | J.5      |
| Serious             | E.2      | C.2      | C.3; I.4   | J.3      | G.2      |
| Minor               |          | G.3; G.4 |            |          |          |
| Negligible          |          | E.5      |            |          |          |

High risk

Moderate risk

Low risk

**Fig 4.** Filtering of HHM risk scenarios in the severity matrix using a threshold level.

**Table II.** Rating and Values—Qualitative

| Rating     | Value |
|------------|-------|
| Low (L)    | 1     |
| Medium (M) | 3     |
| High (H)   | 5     |

decisionmaker's levels of concern. The initial tailoring of the scenarios from HHM to RFRM is subjective. It relies on the expertise of the SMEs that use the methodology. It is only when the scenarios are ultimately selected that the RPN is applied to gain a numeric rating. The completion of this phase results in a more focused subset of risk categories pertinent to the decisionmaker's domain and interest. Referring to the HHM for the electric power grid in Fig. 2, decisionmakers may perform a preliminary filtering

to reduce the dimensionality of the problem. Scenario filtering is invoked depending on the current concerns and priorities of the decisionmakers. Scenario filtering may also be applied to focus on risk scenarios that fall within the purview or responsibility of the decisionmakers. Table I shows the reduced list of risk scenarios that will move forward with the remaining phases of the HHM. For tractability, the ID codes that appear in Column 1 (headtopics) and Column 2 (subtopics) of Table I consistently follow the codes used in the HHM of Fig. 2.

### 5.3. Phase III: Bicriteria filtering and ranking

In this phase, a risk matrix as depicted in Fig. 4 is used to describe the likelihood and consequence levels for the risk scenarios being considered. Tables II–V are used to qualify and quantify the ratings of

**Table III.** Risk Levels of Selected HHM Scenarios against 11 Attributes

| Scenarios Attributes                      | C.2 | C.3 | C.4 | G.2 | H.3 | I.2 | I.4 | J.1 | J.3 | J.5 |
|-------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Undetectability                           | L   | L   | M   | H   | H   | M   | L   | M   | M   | H   |
| Uncontrollability                         | M   | L   | H   | H   | H   | M   | M   | M   | H   | H   |
| Multiple paths to failure                 | M   | M   | H   | M   | M   | M   | M   | M   | H   | H   |
| Irreversibility                           | M   | M   | H   | H   | M   | L   | M   | M   | M   | M   |
| Duration of effects                       | H   | M   | H   | H   | M   | M   | M   | M   | M   | M   |
| Cascading effects                         | M   | M   | H   | H   | H   | M   | L   | M   | M   | H   |
| Operating environment                     | M   | M   | M   | H   | M   | L   | L   | M   | M   | H   |
| Wear and tear                             | M   | M   | L   | L   | L   | M   | L   | L   | L   | L   |
| Hardware/ software/ human/ organizational | M   | L   | M   | M   | H   | M   | M   | H   | M   | H   |
| Complexity and emergent behaviors         | L   | L   | M   | H   | H   | L   | L   | M   | M   | M   |
| Design immaturity                         | L   | L   | M   | M   | M   | L   | L   | H   | H   | H   |

**Table IV.** Impact Rating

| Impact       |   |               |      |
|--------------|---|---------------|------|
| Negligible   | 0 | $\leq a <$    | 1.00 |
| Minor        | 1 | $\leq a <$    | 2.00 |
| Serious      | 2 | $\leq a <$    | 3.00 |
| Critical     | 3 | $\leq a <$    | 4.00 |
| Catastrophic | 4 | $\leq a \leq$ | 5.00 |

**Table V.** Likelihood Rating

| Likelihood | Description                                                         |
|------------|---------------------------------------------------------------------|
| Unlikely   | Very unlikely to occur. Strong Controls in place.                   |
| Seldom     | Not likely to occur. Controls have minor limitations/uncertainties. |
| Occasional | May occur. Controls exist with some uncertainties.                  |
| Likely     | Highly likely to occur. Controls have significant uncertainties.    |
| Frequent   | Nearly certain to occur. Controls have little to no effect.         |

likelihood and impact among the various scenarios. Fig. 4 is assessed by a subjective rating of likelihood versus impact. Previous articles on this subject define the coding of the levels of likelihood and impact both in a quantitative and qualitative sense. The possible intersections of these levels are used to determine the severity of a particular risk scenario. Because the assessment is concerned with the high-severity risk scenarios, those that fall into low-severity categories are set aside. However, the low-severity risk scenarios are not permanently disregarded because they might be of interest in a later RFRM process. The authors recognize that there is a degree of subjectivity in the selection of high-priority scenarios. This

**Table VII.** FMECA Severity and Likelihood Classifications (Francis & Colli, 2014)

| Severity Ranking Criteria          |                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rank - Description</b>          |                                                                                                                                                           |
| 1                                  | Minor failure/degradation, hardly detected, no influence on the system performance                                                                        |
| 2                                  | Failure/degradation will be detected by plant owner/operator and/or will cause slight deterioration                                                       |
| 3                                  | Failure/degradation will be detected by plant owner/operator, will create dissatisfaction, and/or will cause deterioration of parts or system performance |
| 4                                  | Failure/degradation will be easily detected and will cause deterioration or loss of performance                                                           |
| 5                                  | Failure/degradation will result in nonoperation of the system or severe loss of performance                                                               |
| <b>Occurrence ranking criteria</b> |                                                                                                                                                           |
| 1                                  | Unlikely—failure rate per 1/100,000                                                                                                                       |
| 2                                  | Remote probability—failure rate 1/10,000                                                                                                                  |
| 3                                  | Occasional probability—failure rate 1/1000                                                                                                                |
| 4                                  | Moderate probability—failure rate 1/100                                                                                                                   |
| 5                                  | High Probability—failure rate 1/10                                                                                                                        |
| <b>Detection ranking criteria</b>  |                                                                                                                                                           |
| 1                                  | Almost certain that the problem will be detected (chance 81–100%)                                                                                         |
| 2                                  | High probability that the problem will be detected (chance 61–80%)                                                                                        |
| 3                                  | Moderate probability that the problem will be detected (chance 41–60%)                                                                                    |
| 4                                  | Low probability that the problem will be detected (chance 21–40%)                                                                                         |
| 5                                  | None/minimal probability that the problem will be detected (chance 0–20%)                                                                                 |

subjectivity is necessary in phase 2 for the stakeholders/ policymakers to scope the problem using subject matter expertise. It also shows the current state of affairs as the ultimate goal is a transition to the proposed model and a “quantifiable” value through the use of FMECA once the initial triage of the situation

**Table VI.** List of Six Scenarios to be Evaluated in Phases V and VI

| Headtopics              | Subtopic ID | Subtopic Name     | Failure Mode                      | Mitigation Factors                                 |
|-------------------------|-------------|-------------------|-----------------------------------|----------------------------------------------------|
| C. Physical             | C.4         | Generation Plants | Malicious code inject             | Restrict device access (both physical and logical) |
| G. Human                | G.2         | Insider Threat    | Employee mass remote disconnect   | Use Role-Based Access Control (RBAC)               |
| H. Management           | H.3         | Security          | Weak encryption exposes devices   | Require approved cryptographic algorithms          |
| J. System Configuration | J.1         | SCADA             | SCADA issues invalid commands     | Authenticate users accessing the DER SCADA system  |
| J. System Configuration | J.3         | Network           | Inadequate Network segregation    | Require intrusion detection and prevention         |
| J. System Configuration | J.5         | Software Updates  | Known but unpatched vulnerability | Implement configuration management                 |

is complete. Indeed, risk assessment can include uncertainty or a range of likelihood/impact.

#### 5.4. Phase IV: Multicriteria Filtering

This phase requires the use of 11 criteria to evaluate the impact of the risk scenarios against the defensive abilities of the underlying system. The impact on the system of each risk scenario is judged high (H), medium (M), and low (L) according to the 11 attributes. Table VI shows a sample chart which evaluates the filtered risk scenarios from Phase III against 11 attributes of the system's defense capabilities. A more rigorous discussion of the 11 attributes is found in Haimes et al. (2002). This process enables prioritizing the risk scenarios and will inform the process in Phase V. Thus, Phase V is executed to further reduce the number of risk scenarios into a more manageable size, in light of information from Phase IV and additional inputs from experts.

The qualitative filtering has left us six remaining scenarios that will be subjected to a quantitative ranking through the use of FMECA. The "threshold" for the scenarios to be further assessed and managed is delineated by the scenarios in the high-risk category. The list of scenarios is depicted by Table VI.

#### 5.5. Phase V: Quantitative Ranking

In our set of scenarios, FMECA is used to quantify the results after the filtering process. The RPN is used to conduct the risk assessment. The three factors are all scored from 1 (best) to 5 (worst) on the basis of degree. RPN is the product of occurrence, detection, and severity, which is expressed as  $RPN = \text{Severity (SEV)} \times \text{Occurrence (OCC)} \times \text{Detection (DET)}$ .<sup>(46)</sup> This simple, yet powerful formula allows us to rank the possible threats against the system, and additionally allows us to develop mitigation courses of action, as well as cost-benefit analysis of these events. The case study of the electric grid is portrayed in a FMECA Worksheet Analysis in Table VII. A simplified version is displayed below. Applying the basic formula, we can develop the necessary numbers to consider the probability and severity of each discrete event. Though subjective, these RPNs give us a starting point from which to develop a holistic strategy to defend against cyber-attacks on the electric grid.

Using the scales in Table VII to quantify the severity, occurrence, and detection of the remaining subcategories, we begin to develop the most lethal

of scenarios for cyber-attack. The FMECA Analysis Worksheet Excerpt (Table VIII) quantifies the RPNs.

Examples of individual SME ranking of the scenarios are given in Figs. 5–8. Some of the SMEs ranked the scenarios as discrete events while others considered some of the scenarios to be equal in measure for the RPN. The risk of relying on one expert, with their associated biases, becomes starkly evident as the graphs depict radically different representations to the same set of rankings.

Table VIII shows the FMECA Analysis Worksheet Excerpt for SME1. Conversely, Fig. 9 shows the smoothing effect of the combined individual graphs both in numeric and graph form. This illustrates the importance of convening a "focus group" of SMEs when faced with a situation as opposed to relying on one particular SME.

#### 5.6. Phase VI: Risk Management

After Phase V, only a few risk scenarios remain. The aim of Phase VI is to generate several options to reduce, if not eliminate the risk scenarios. A cost-benefit analysis is the heart of phase VI. Cost, risk reduction, and other options for management of the most critical of the scenarios are performed at this stage. This stage is key for stakeholders and policy-makers, as each of the scenarios has an associated cost, likelihood, and consequence, which are weighed against the resiliency and robustness of the system.

#### 5.7. Phase VII: Safeguarding against Missing Critical Items

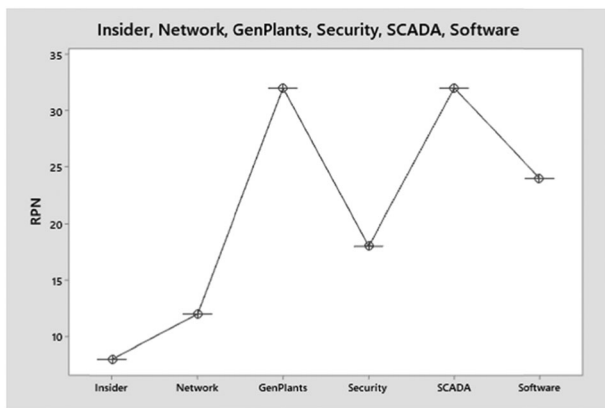
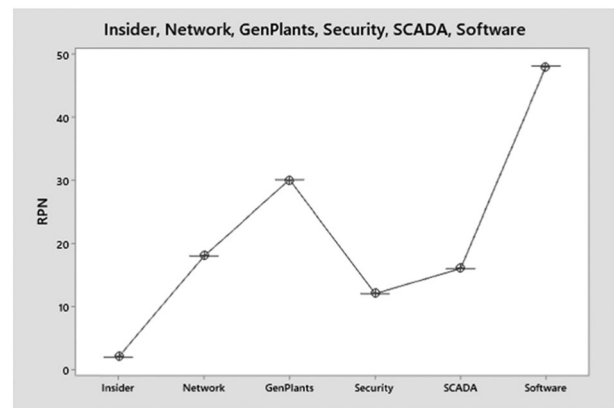
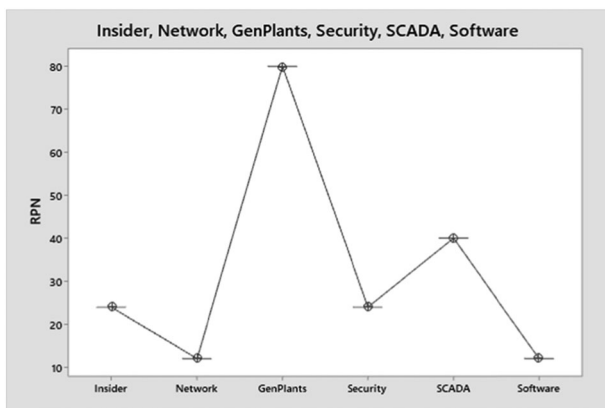
The entire system is examined by taking into account the risk management policies identified in the previous phase. The effectiveness of these policies is reevaluated by considering the additional impacts of the filtered-out risk scenarios. This provides insight into a number of alternative management options that otherwise might have been overlooked. The survey offers insight into some key areas about the risk management policies currently in place. The survey represents both the U.S. population and the expanded SME group.

The results of our survey offered both intuitive and counterintuitive responses. Sample comments from the survey are listed in Table IX.

One of the most telling of the series of survey questions points to the fact that the preparation for a cyberattack is different than the preparation

**Table VIII.** FMECA Analysis Worksheet Excerpt for SME1

| Sub-Topic ID        | Scenario                  | Potential Failure Mode            | Severity                  | Occurrence           | Detection                  | Risk Priority Number (RPN)             |
|---------------------|---------------------------|-----------------------------------|---------------------------|----------------------|----------------------------|----------------------------------------|
| Alpha- numeric Code | Larger Scenario Presented | Failure exhibited                 | How severe is the effect? | How likely to occur? | How probable is detection? | Risk Priority Number (SEV) (OCC) (DET) |
| G.2                 | Insider Threat            | Employee mass remote disconnect   | 2                         | 1                    | 4                          | 8                                      |
| J.3                 | Network                   | Inadequate Network segregation    | 4                         | 3                    | 1                          | 12                                     |
| C.4                 | Generation Plants         | Malicious code inject             | 4                         | 2                    | 4                          | 32                                     |
| H.3                 | Security                  | Weak encryption exposes devices   | 3                         | 3                    | 2                          | 18                                     |
| J.1                 | SCADA                     | SCADA issues invalid commands     | 4                         | 2                    | 4                          | 32                                     |
| J.5                 | Software updates          | Known but unpatched vulnerability | 3                         | 4                    | 2                          | 24                                     |

**Fig 5.** SME1.**Fig 7.** SME3.**Fig 6.** SME2.

for a natural disaster. The archives are filled with data from the previous responses to tornadoes, floods, and hurricanes. However, the need to capture the “lessons learned” from cyberattacks is only beginning. The responses to the survey questions in Figs. 10 and 11 point to a need to examine the preparations currently in place.

A significant percentage of respondents, from both groups, agree that the government would prepare differently for a cyberattack.

## 5.8. Phase VIII: Operational Feedback

This phase looks at the methodology in terms of the dynamic nature of risk assessment and

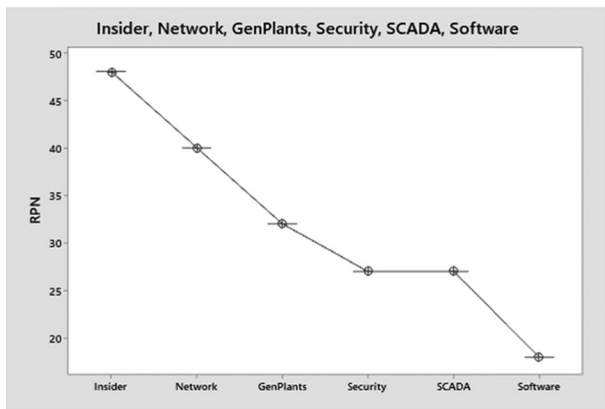


Fig 8. SME4.

management. The experience and information gathered from the previous phases of RFRM helps refine and update the prioritization of the risk scenarios and improves the decision-making process.

Downed power lines, destroyed substations, and substantial damage to major power generation plants through flooding or other damage caused by gale force winds is the major threat. In addition, this damage is more readily visible to the naked eye and can be quickly assessed and in most cases rapidly repaired. Conversely, the damage to the electric grid

from cyber-attack may be more insidious and more difficult to repair.

This area has not been previously explored especially in the context of cyber-attacks on the electric grid. In general, the loss of functionality of components of the electric power grid must be recovered. The lost capability must be recovered no matter the cause—natural disaster or cyber-attack. In addition, natural disasters offer more historical observations of the response to the phenomena. Therefore, the knowledge transfer from natural disasters to cyber-attacks acts as a good proxy for modeling such events.

Natural disasters may cause more physical damage to the electric grid than cyber-attacks. However, while cyber-attacks may exhibit less physical damage the degradation of the system may be worse. An example of this would include malware left behind by an attacker that could be activated at a date and time of their choosing. For example, the failure of the electric grid could stem from several independent acts and actors rendering system failure a consequence that must quickly be rectified in order to return normal operations. Such scenarios could be revisited through the operational feedback phase of the RFRM. Evidently, the type of risks and their complexities evolve over time. Hence, it is important to implement the proposed framework as a continuous process to capture

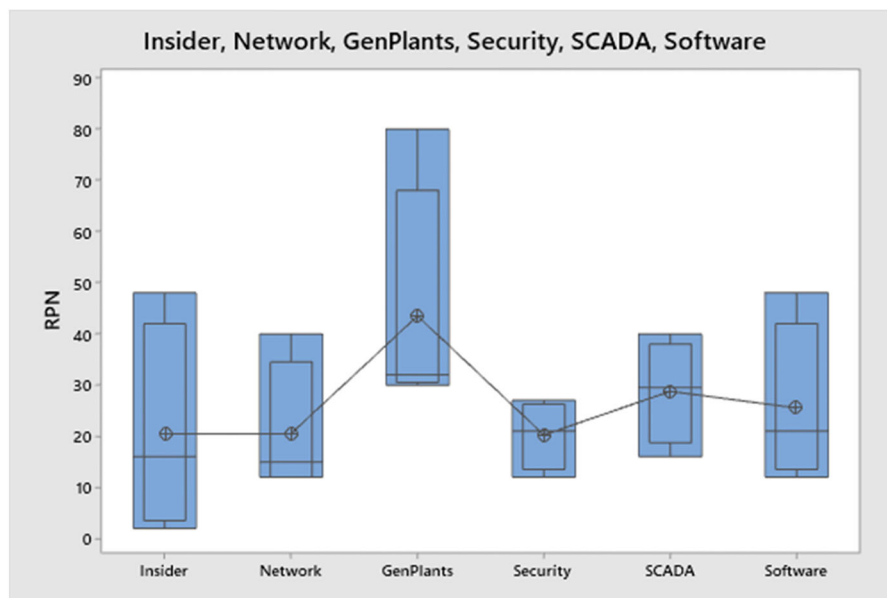


Fig 9. Boxplot graph of combined SMEs.



**Table IX.** Sample Comments from the Survey

| Survey Question                                                                                                                        | Comment                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cyberattack on the electric power grid is likely to occur in the near future.                                                          | I believe the loss of electrical power is one of the largest and most underestimated threats of the modern world.                                                                                     |
| Current policies would help to mitigate a cyberattack on the electric power grid.                                                      | From what I have read and heard; we do not have enough protection of our power grid.                                                                                                                  |
| Current policies would help to mitigate a cyberattack on the electric power grid.                                                      | Strongly disagree. The grid is very vulnerable and can be taken down with ease by other countries.                                                                                                    |
| I would expect the cascading effects of a cyberattack on the electric power grid to be greater on the economy than a natural disaster. | Clearly, a coordinated cyberattack would be more dangerous and debilitating than even the largest of possible natural disasters because cyberattack would likely interfere with the defense response. |
| I believe that we are better able to mitigate (with the current policies in place) the effects of a natural disaster or cyberattack.   | Cyberattacks on infrastructure are a HUGE risk that we definitely need to be doing more to mitigate.                                                                                                  |

the changing risk landscape of the electric power grid.

## 6. EPILOGUE

This article presented a framework for risk analysis for cyber-security and critical infrastructure protection of the electric power grid. The framework aims to link HHM, RFRM, and FMECA as a structure to qualitatively and quantitatively address the risk associated with the operation of the electric power grid.

Returning to the questions posed by Haimes (1991):

- What can be done and what options are available?
- What are the trade-offs regarding costs, benefits, and risks?
- What are the impacts of current decisions on future options?

Many studies have shown that outages of the electric grid can have enormous economic effects on the U.S. economy. However, these studies have predominantly focused on the effects of discrete, short-term natural disasters on the electric grid.

Many areas still need exploration, especially in cyber-security. Natural disasters events offer valuable lessons. However, cyber-attacks represent a clear point of departure for a more complex series of problems.

This article reveals several areas for further discussion and examination. There is a need to recognize the current electric grid's cyber current vulnerabilities. We also need to better understand the effects of a coordinated cyber-attack on the American electric grid. The major agencies across the government need to develop strategies for a coordinated response to such an event. The recently released Energy Sector Cybersecurity Framework Implementation Guidance is groundbreaking in its recognition of risk management tools, processes, standards, and guidelines (Department of Energy, 2015). In addition, scenarios, exercises, and training are needed to practice the response and glean new knowledge that should be codified in policies and procedures to react to this type of events. The desire to "operationalize" this concept is real and important. Discussions have been had with the Cyber Security Infrastructure Security Agency (a newly created agency in the Department of Homeland Security) to consider ways to "operationalize" the risk associated with a cyber-attack on the electric grid. This endeavor has been undertaken in order to allow policymakers, stakeholders, and other parties involved to triage the cyber event and to make real-time decisions for strategies, resources allocation, while fundamentally understanding the inherent risk associated with the possible alternatives.

The fundamental questions of risk assessment (Kaplan & Garrick, 1981) and risk management (Haimes, 1991) form the cornerstone of the iterative process necessary to define the limits of the cyber-attack problem set. Risk analysis and risk management techniques are key to asking the right questions to plumb the depths of a synchronized response to a reality of today's world. Natural disaster strategies can be used as a starting point, but a cyber-attack on the electric grid represents a new, more dangerous challenge. Coordinated efforts at all levels of government are key to developing strategies, policies, and procedures to minimize the disruption that these events potentially represent. Finally, new avenues need to be explored to gain the benefits of public-private partnership, as part of the response to this new and evolving threat. In addition, further development of cyber-security standards is needed to provide a baseline to protect

Q4 The United States Government would prepare differently for a cyber-attack on the electric power grid vice a natural disaster.

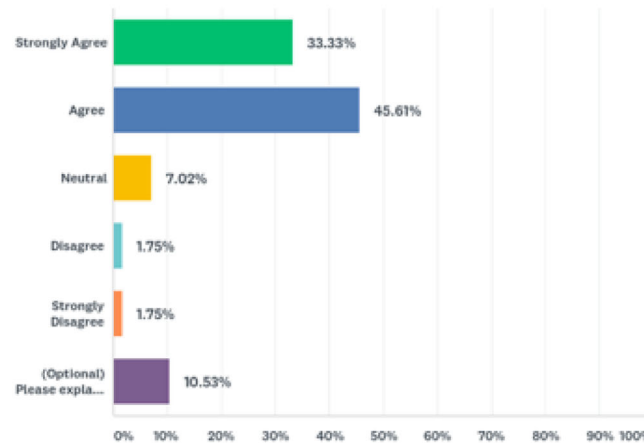


Fig 10. SME population.

Q4 The United States Government would prepare differently for a cyber-attack on the electric power grid vice a natural disaster.

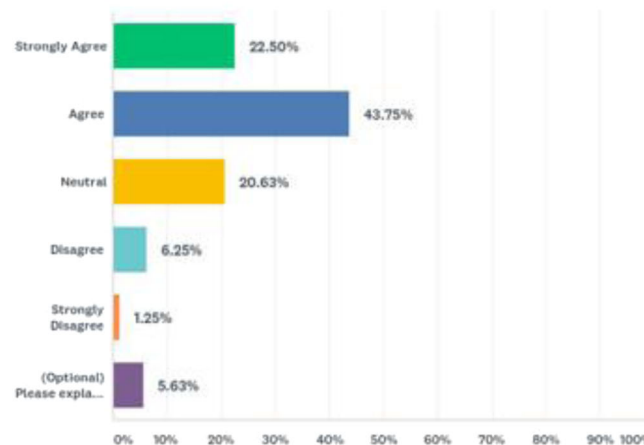


Fig 11. Cross-section of US population.

against known vulnerabilities, while advancing risk management strategies to improve decision making.

Ted Koppel's book "*Lights Out: A Cyberattack, a Nation Unprepared, Surviving the Aftermath*" (2015) cites the increasing risks associated with a coordinated cyber-attack on the American electric grids. In addition, works by James Lewis of the Center for Strategic and International Studies and others recognize the growing threat against the network. The vast majority of current studies center on disaster recovery of the grid from a natural event. The

analysis of cyber-security risks to the American electric grid remains largely unexplored. The increasing awareness of the catastrophic threat to the homeland because of an attack on the electric grid has reinforced the requirement for a coordinated governmental approach. The disastrous risks, due to the *ad hoc* nature of the current state of cyber security in the United States, necessitate the need to synchronize the internal efforts of the U.S. Government with academic and research institutions that are involved with the development of disaster risk assessment and



management models for critical infrastructure systems, such as the electric power grid.

Recognizing the vastness of the attack surface of the electric grid and the economic incentives this work allows stakeholders and policymakers to triage the immensity of data in order to attempt to quantify their recommendations for actions and resources. This framework can be applied at the strategic, operational, and tactical levels of decision making. The advantage of the use of this model at all levels allows for a common basis of understanding for the policymaker and decisionmaker alike as they have common reference “language” as they traverse different levels and with the different agencies involved in the problem set.

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